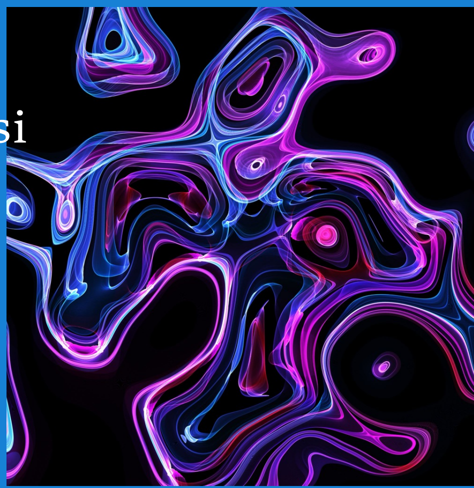
II 1200
Pengantar
Sistem dan Teknologi Informasi



Minggu 11
Enterprise Resource Planning

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School of Electrical Engineering and Informatics, ITB

Semester 2 2025-2026

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TUJUAN PEMBELAJARAN

- Menjelaskan konsep dasar metodologi pengembangan sistem informasi no-code
- Mengidentifikasi tahapan-tahapan pengembangan.
- Menjelaskan manfaat dan penggunaan platform No-Code seperti Odoo.
- Memahami bahwa SI mencakup aplikasi, data, proses, infrastruktur, organisasi, dan brainware.

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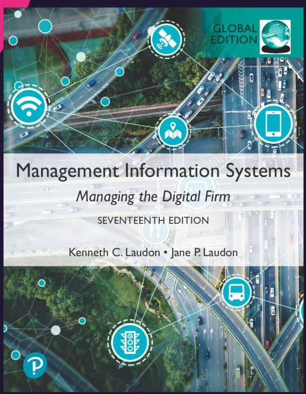
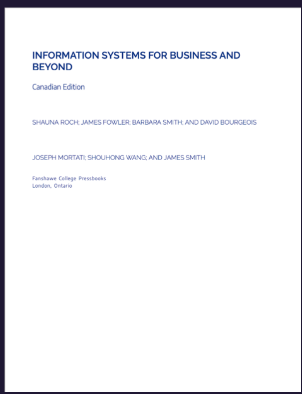
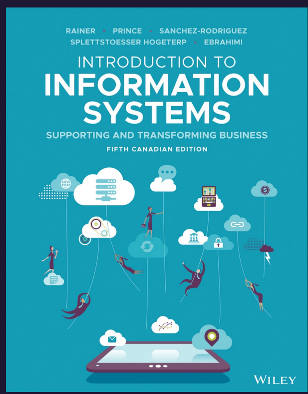
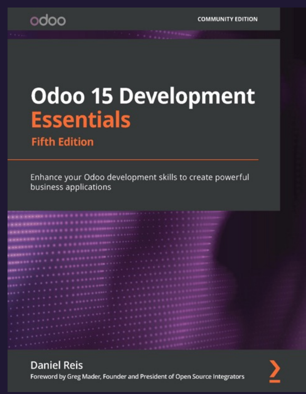
REFERENSI









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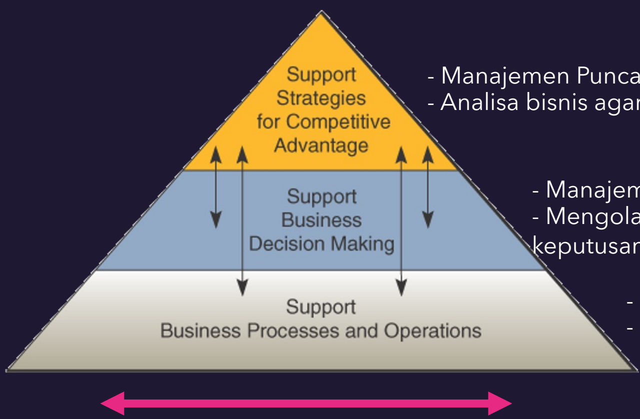
PERAN DASAR SISTEM INFORMASI



- Manajemen Puncak
- Analisa bisnis agar bisa berkompetisi

- Manajemen menengah
- Mengolah pelaporan, analisis untuk pembuatan keputusan

- Para Pelaksana
- Mengolah transaksi operasi

Fungsional

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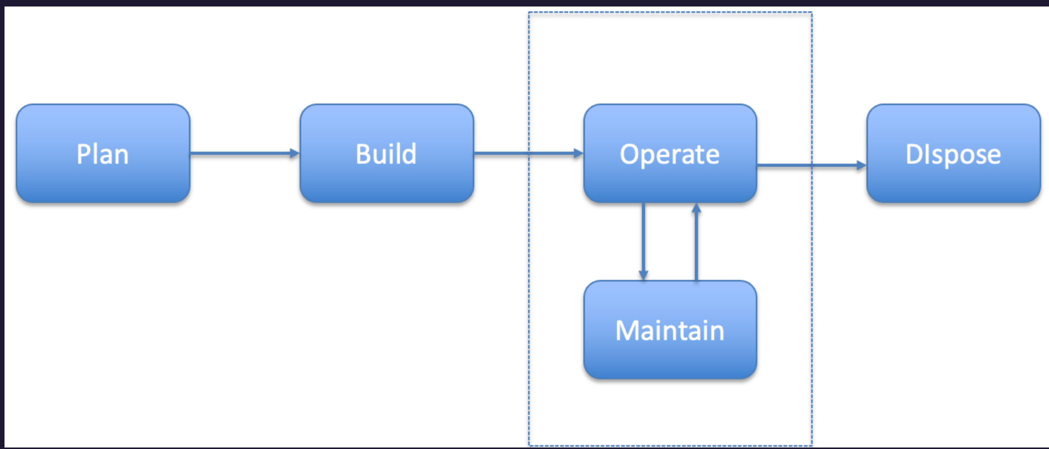
SIKLUS HIDUP SISTEM INFORMASI (SYSTEM LIFECYCLE)



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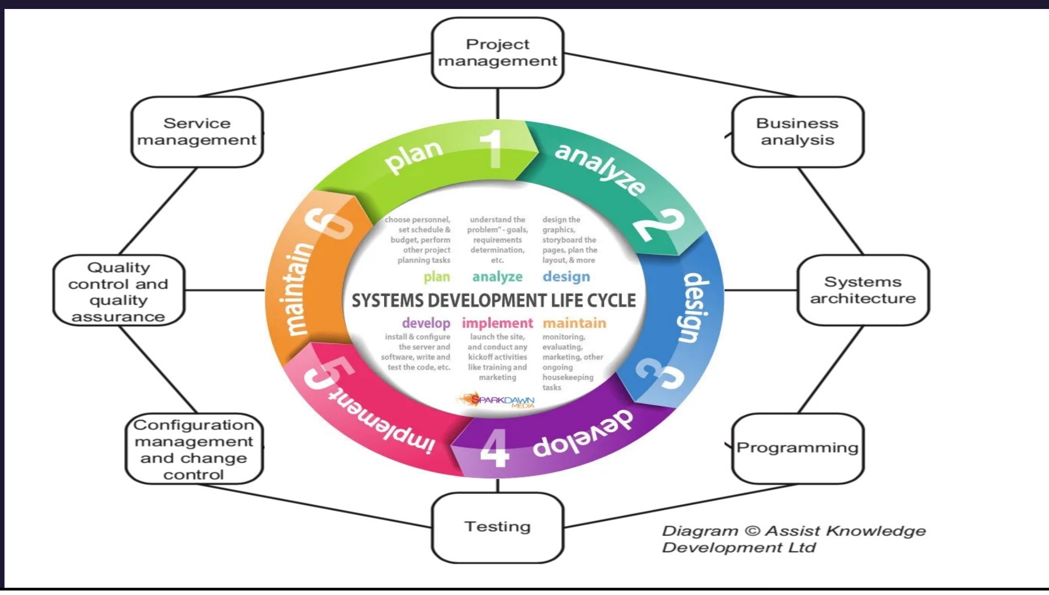
System Development Life Cycle (SDLC)



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TAHAPAN UMUM SDLC

1. **Perencanaan:** Tujuan proyek & sumber daya.
2. **Analisis:** Kebutuhan pengguna dan bisnis.
3. **Desain:** Struktur sistem & tampilan antarmuka.
4. **Pengembangan:** Pemrograman & instalasi sistem.
5. **Pengujian:** Cek apakah sistem berfungsi.
6. **Implementasi:** Pindah ke sistem nyata.
7. **Pemeliharaan:** Perbaikan & pembaruan rutin.

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KOMPONEN SI dalam SDLC

- Proses Bisnis: alur kerja terintegrasi.
- Aplikasi: alat bantu pemrosesan data.
- Data: sumber informasi utama.
- Infrastruktur: jaringan, server.
- Organisasi: struktur dan peran.
- Brainware: pengguna dan pengembang

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1. TAHAP PERENCANAAN

- Mengidentifikasi tujuan, ruang lingkup, sumber daya, dan jadwal proyek.
- Menetapkan ruang lingkup sistem (global)
  - Fungsionalitas sistem
  - Struktur dan aliran data
  - Alur kerja dan proses bisnis
  - Infrastruktur
  - Struktur organisasi pendukung
  - Peran & kompetensi pengguna/ pengelola

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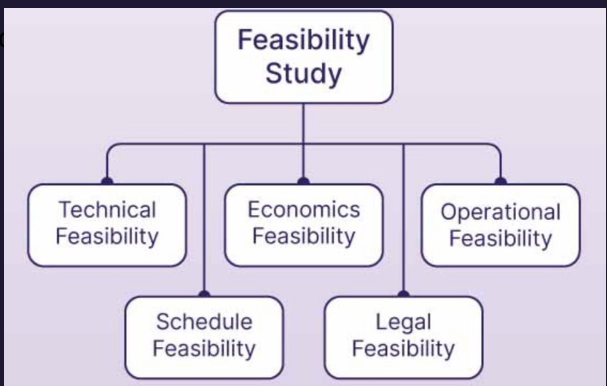
Tahap Perencanaan: **Studi Kelayakan**

- Time c

Can we build it?

Time constraint?



Will users accept it?

Against any rulles?

Economic Feasibility

- Total cost of ownership (TCO)
- Tangible benefits
- Intangible benefits

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2. TAHAP ANALISIS

Definisi

- Fokus pada **identifikasi kebutuhan** sistem dari perspektif pengguna dan bisnis.
- Tujuan utama adalah **memahami apa yang harus dilakukan** oleh sistem agar sesuai dengan ekspektasi pengguna.
- Analisis mencakup eksplorasi kebutuhan terhadap aplikasi, data, proses, organisasi, infrastruktur, dan brainware.

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3. TAHAP DESAIN

Definisi

- Untuk menerjemahkan kebutuhan sistem menjadi solusi teknis.
- Tahapan ini menjembatani kebutuhan pengguna dengan rancangan sistem yang dapat dikembangkan oleh tim teknis.
  - Memastikan kebutuhan dan peran pengguna terakomodasi dalam desain.
- Desain mencakup aplikasi, struktur data, proses bisnis, infrastruktur pendukung, tata kelola organisasi, dan peran pengguna.

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4. TAHAP PENGEMBANGAN

Definisi

- Rancangan sistem diubah menjadi sistem yang nyata dan dapat dijalankan.
- Fokus pada penerapan teknis terhadap desain sistem menggunakan bahasa pemrograman dan alat bantu pengembangan.
- Tidak hanya mencakup pengembangan aplikasi, tetapi juga pengelolaan data, infrastruktur, proses bisnis, organisasi, dan pelibatan brainware.

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5. TAHAP PENGUJIAN

Definisi

- Memverifikasi dan memvalidasi sistem agar sesuai dengan kebutuhan yang telah ditentukan.
- Tujuannya adalah memastikan sistem bekerja dengan benar, bebas dari kesalahan kritis, dan siap digunakan oleh pengguna akhir.
- Tahap ini mencakup pengujian terhadap aplikasi, validasi data, pengujian proses bisnis, serta kesiapan infrastruktur, organisasi, dan brainware.

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6. TAHAP IMPLEMENTASI

Definisi

- Memasang (Deploy) sistem baru dan mulai menggunakan (Live-On) dalam lingkungan operasional.
- Fokus utama adalah memastikan sistem dapat berjalan dengan baik dan pengguna siap mengoperasikannya.
- Implementasi mencakup aspek aplikasi, data, infrastruktur, organisasi, dan brainware.

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7. TAHAP PEMELIHARAAN

Definisi

- Memastikan sistem tetap berjalan optimal setelah implementasi.
- Mencakup pembaruan sistem, perbaikan, dan penyesuaian terhadap kebutuhan baru.
- Melibatkan pengelolaan aplikasi, data, infrastruktur, dan pengguna

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TAHAP PEMELIHARAAN

Risiko

- Tidak tersedianya tim atau anggaran pemeliharaan.
- Perubahan sistem dilakukan tanpa standar (tidak terdokumentasi).
- Ketidaksesuaian antara pembaruan sistem dan proses organisasi.

Kunci Keberhasilan

- Tersedianya mekanisme pelaporan dan pemrosesan perubahan.
- Perawatan menyeluruh terhadap aplikasi, infrastruktur, dan data.
- Keterlibatan pengguna dalam evaluasi sistem.
- Siklus pembelajaran berkelanjutan melalui pelatihan dan umpan balik.

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Metodologi Pengembangan

Metodologi	Karakteristik Utama	Kelebihan	Kekurangan	Cocok Untuk	Tidak Cocok Untuk
Waterfall	Linear, berurutan, dokumentatif	Terstruktur, mudah dimanajemen, dokumentasi kuat	Tidak fleksibel, feedback terlambat	Sistem keuangan, ERP, proyek regulatif	Proyek dengan perubahan cepat, startup
Agile	Iteratif, kolaboratif, fleksibel	Fleksibel, user-centric, hasil cepat terlihat	Dokumentasi bisa kurang, perlu komitmen tinggi	Aplikasi modern, sistem startup, sistem digital	Organisasi birokratis, proyek formal
Prototyping	Fokus pada validasi awal melalui model visual	Kurangi miskomunikasi, cocok untuk UI/UX	Tidak cocok untuk sistem kompleks, dokumentasi bisa kurang	Portal informasi, sistem publik, UI intensif	Backend kompleks, sistem tanpa UI langsung
RAD	Cepat, modular, menggunakan alat bantu visual	Iterasi cepat, cocok untuk digitalisasi proses bisnis	Tidak cocok untuk sistem besar dan kompleks	Sistem administratif, layanan digital organisasi	Sistem skala besar dan kritikal
No-Code	Visual, tanpa coding, cocok untuk pengguna non-teknis	Cepat dikembangkan, mudah digunakan non-programmer	Fungsionalitas terbatas, bergantung pada platform	Digitalisasi proses kecil, pelaporan komunitas	Sistem kompleks, skalabilitas tinggi

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NO (Minimal) CODE

- Metodologi Pengembangan dengan meminimalkan pemrograman.
  - Menggunakan paket perangkat lunak
  - Biasanya berupa Sistem Enterprise seperti Sistem ERP (Enterprise resource Planning)
  - Karena proses bisnis Perusahaan berbeda dengan proses bisnis ERP akan ada penyesuaian!

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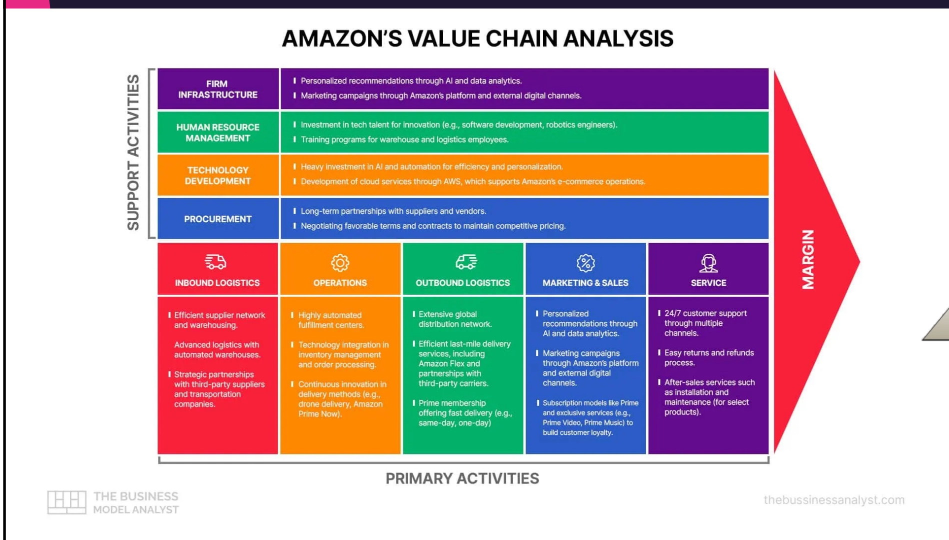
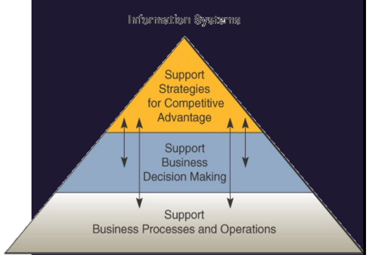
Masih Ingat Value Chain?





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Dulu akan banyak Sistem Informasi

- Sistem Informasi Keuangan
- Sistem Informasi Kepegawaian
- Sistem Informasi Pengadaan
- Sistem Informasi Pergunakan (Logistik)
- Sistem Informasi Operasi
- Sistem Informasi Distribusi
- Sistem Informasi Pemasaran dan penjualan
- Sistem Informasi Layanan Konsumen

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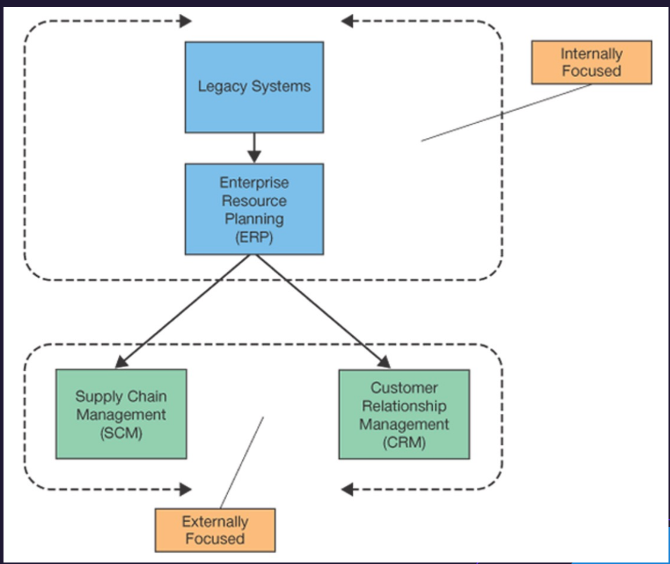
SISTEM ENTERPRISE



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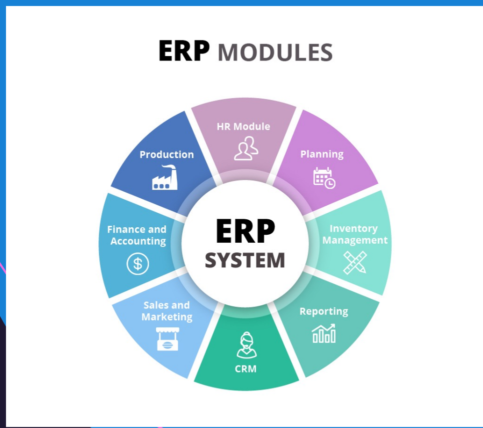
SISTEM ENTERPRISE: ERP

- Enterprise Systems
  - Enterprise Resource Program
  - Enterprise Resource Planning Software
  - Enterprise Resource Planning System
- An integrated system consisting of integrated applications with common database which coordinate business activities and support the flow of information across the enterprise.
- Includes financial, operational and strategic systems



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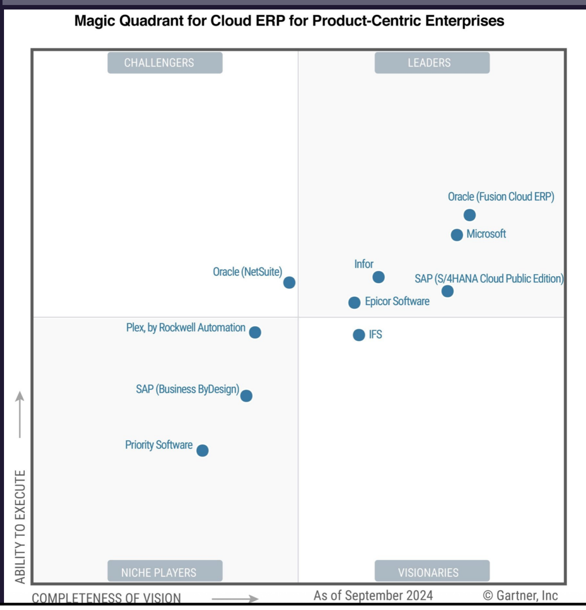
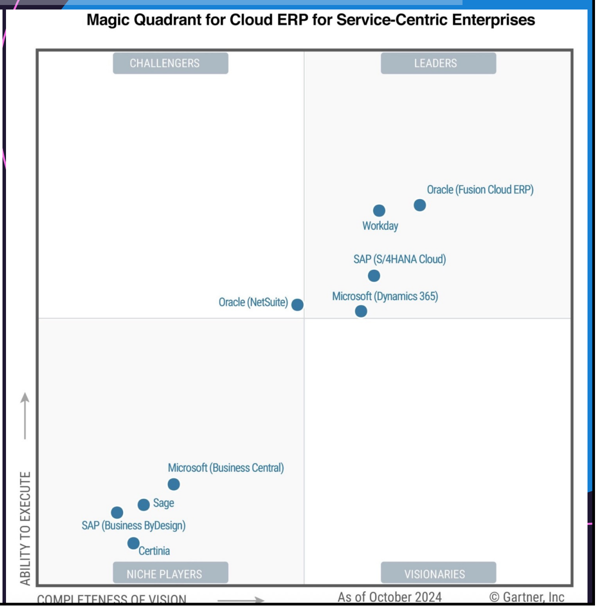
SISTEM ENTERPRISE: ERP





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MENGAPA ERP

Value Category	Value Lever	Feasible Improvement Range	Sales & Operations Planning	Order Fulfilment	Operations & Logistics	Finance & Admin.	Maint. & Supply Sourcing
Distribution	Reduced transportation cost	5 - 30%	◐	○	●	○	○
	Reduced distribution cost	5 - 20%	◐	○	●	○	○
Labor	Reduction in Order Fulfillment labor cost	10 - 25%	◐	●	○	○	○
	Reduction in Finance & Admin. labor cost	10 - 30%	○	○	◐	●	○
	Reduction in Maintenance & Sourcing labor cost	5 - 20%	○	○	○	○	●
Working Capital	1-Time reduction in FG inventory	10 - 35%	●	◐	○	○	○
	1-Time reduction in MRO inventory	5 - 20%	○	○	○	○	●
	1-Time reduction in WIP inventory	1 - 10%	●	○	○	○	○
	1-Time reduction in AR DSO	1 - 10%	○	●	◐	○	○
COGS	Reduced procured/direct material cost	1 - 5%	○	○	○	○	●
	Reduced inventory carrying cost	10 - 30%	●	○	○	○	○
	Reduced scrap and rework cost	5 - 20%	●	○	○	○	○
Revenue/ Margin	Improved mix/price/resource use	1 - 3%	◐	○	◐	◐	○
	Increase volume via recovery of lost sales	1 - 3%	◐	◐	◐	○	○
	Improved customer service	1 - 2%	◐	◐	◐	◐	○

Source: Accenture global survey 1999 – Natural Resources clients

Shading indicates degree of relevance expressed by respondent (no fill = low) 10

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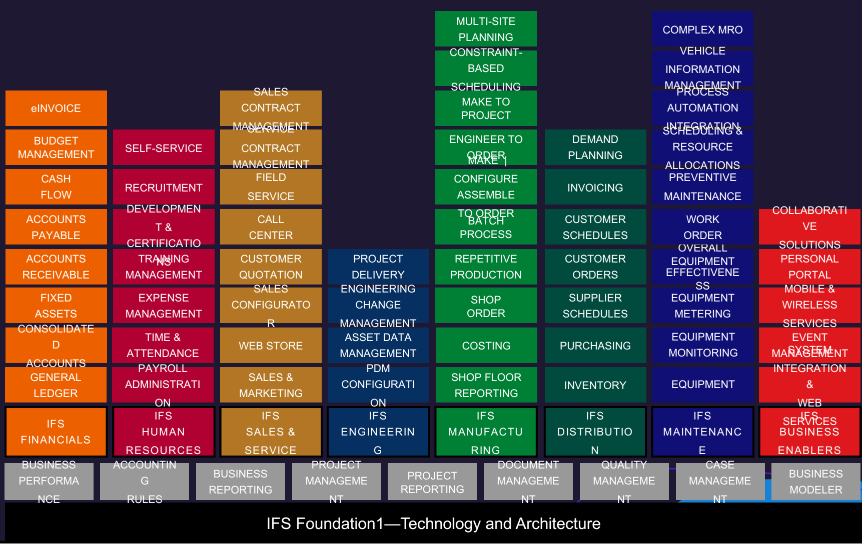
SISTEM ERP: IFS

- Modules: Choose what we need



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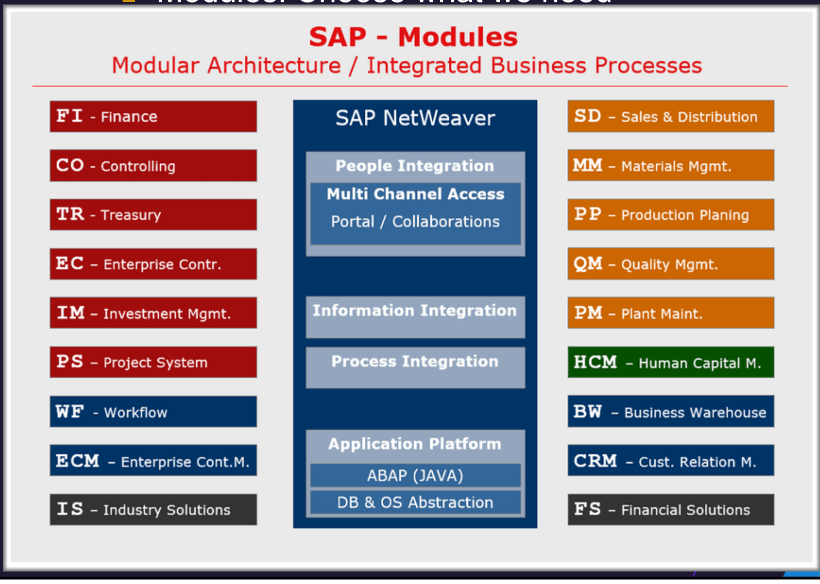
SISTEM ERP: SAP

- Modules: Choose what we need



Accounting

- SAP-FI - Financial Accounting
  - FI-GL: General Ledger
  - FI-AP: Accounts Payable
  - FI-AR: Accounts Receivable
  - FI-AA: Asset Accounting
  - FI-BL: Bank Ledger
  - FI-SL: Special Purpose Ledger
  - FI-TV: Travel Management
- SAP-TR - Treasury
- SAP-RE - Real Estate Management
- SAP-EC - Enterprise Controlling
- SAP-IM - Investment Management
- SAP-PS - Project System
- SAP-CO - Controlling
  - CO-OM: Overhead Cost
  - CO-PC: Product Cost
  - CO-PA: Profitability Analysis
  - EC-PCA: Profit Center Accounting

Logistics

- SAP-MM - Materials Management
- SAP-SD - Sales & Distribution
- SAP-PP - Production Planning & Control
- SAP-QM - Quality Management
- SAP-LO - Logistics General
- SAP-PM - Plant Maintenance
- SAP-LE - Logistics Execution
- SAP-CS - Customer Service
- SAP-EHS - Environment, Health & Safety
- SAP-SRM - Supplier Relationship Manag.
- SAP-PLM - Product Lifecycle Management

Supply-Chain-Management

The term ***Supply Chain Management*** refers to the cross-company (from the supplier to the customer) coordination of the material, financial and information flows across the entire value chain.

- SAP-SCM - Supply Chain Management
- SAP-APO - Advanced Planning & Optimization

Human Resources

- SAP-HCM - Human Capital Management (formerly: HR - Human Resources)
- ESS - Employee Self-Services
- MSS - Manager Self-Services

Cross Application Modules

- SAP-WF - Workflow
- SAP-BW - Business Warehousing; DWH
- SAP-CRM - Customer Relationship Manag.
- SAP-ECM - Enterprise Content Management
- SAP-MII - SAP Manufacturing Integration & Intelligence

Technical Modules

- SAP NetWeaver (formerly SAP Basis)
- SAP Solution Manager
- SAP-EP - Enterprise Portal
- SAP-PI - Process Integration (formerly SAP-XI)

Industry Modules

These SAP Modules employ the prefix "IS" for "Industry Solutions".

- IS-A - Automotive
- IS-H - Healthcare
- IS-M - Media
- IS-PS - Public Sector
- IS-R - Retail
- IS-T - Telecommunication
- IS-U - Utilities
- IS-RE - Real Estate
- IS-OIL - Oil & Gas

These SAP Modules employ the prefix "FS" for "Financial Services" (Insurance Companies & Banks).

- FS-BP - Business Partner
- FS-CMS - Collateral Management
- FS-CML - Consumer, Corp. Mortgage Loans
- FS-CM - Claim Management
- FS-PM - Policy Management
- FS-CD - Collection & Disbursement
- FS-ICM - Incentive / Commission Management
- FS-RI - Reinsurance

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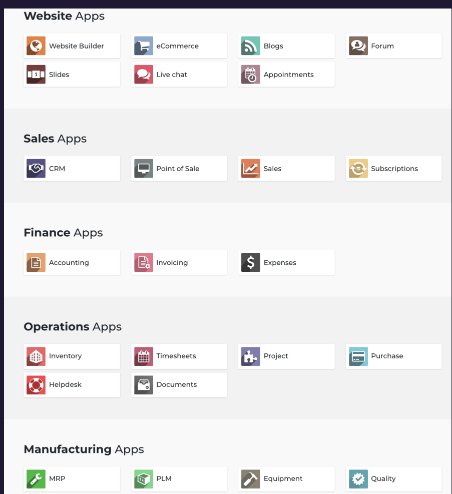
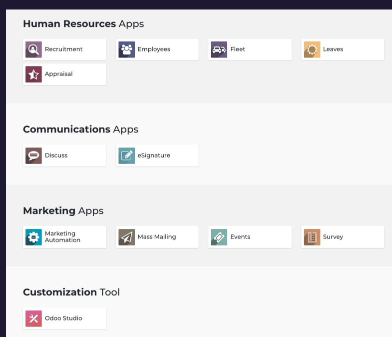
SISTEM ERP: ODOO

- Modules: Choose what we need





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STRATEGI IMPLEMENTASI

- "Vanilla" version
- Customizations
- Best practices
- Business process reengineering (BPR)

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SISTEM ENTERPRISE: CRM

- Customer Relationship Management (CRM)
  - Sales Force Automation (SFA)
  - New opportunities for competitive advantage
  - Examples:
    - MGM
    - American Airlines
    - Marriott International

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SISTEM ENTERPRISE: SCM

- Supply Chain Management (SCM)
  - Supply chain – the producers of supplies that a company uses
  - Supply network
  - What if supply chain does not collaborate?
  - Two objectives of upstream information flow:
    - Accelerate product development
    - Reduce costs associated with suppliers

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SDLC untuk ERP

Traditional SDLC Phase	Holland & Light ERP Phase	Key Activities in ERP Context
1. Planning	**Initiation**	- Define ERP goals and strategic alignment - Conduct feasibility analysis - Secure sponsorship and form team
2. Analysis	**Adoption**	- ERP vendor selection - Fit-gap analysis - Establish project scope and governance
3. Design	**Adaptation**	- Configure ERP to match business processes - Redesign (BPR) where needed - Plan infrastructure and data integration
4. Development	**Adaptation** *(continued)*	- Setup ERP modules - Customize reports, interfaces (if needed) - Prepare data migration scripts
5. Testing	**Acceptance**	- Conduct unit/integration testing - User Acceptance Testing (UAT) - Final validation and dry runs
6. Implementation	**Routinization**	- System go-live - Rollout to users - Transition from legacy to ERP processes
7. Maintenance	**Infusion**	- Post-live support - Feature optimization and upgrades - Continuous process improvement

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Fit-Gap Analysis

1. **Document current (as-is) business processes**
   - How tasks are currently performed in departments (e.g., procurement, finance, HR).
   - This is done through **interviews, workshops, observations**, and reviewing SOPs.
2. **Identify target (to-be) processes or business goals**
   - What the organization wants to achieve post-ERP.
   - This often involves identifying pain points in current workflows.
3. **Map ERP capabilities to these processes**
   - Check if the ERP modules can handle the needed processes.
   - If there's no match, identify the **gap** and decide whether to:
     - **Reengineer** the business process to fit the ERP ("process alignment"), or
     - **Customize** the ERP (usually discouraged due to cost and complexity).

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Key Considerations

🧠 Key Considerations

Criteria	Build Custom	Buy ERP
Customization	Fully customizable to unique needs	Limited to ERP's architecture and config tools
Cost (Initial)	Often higher due to dev & testing costs	Lower (but license fees can be high)
Cost (Long-term)	Maintenance, updates, scalability handled internally	May increase with annual fees and upgrades
Implementation Speed	Slower; dev time required	Faster; off-the-shelf readiness
Vendor Lock-in	No (unless outsourced development)	High (ERP dependency and upgrade cycles)
Process Standardization	May keep existing ad-hoc processes	Pushes best practices through standard modules

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